

GANESH HEGDE, PhD

Energy Systems · Climate Policy · Development

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RESEARCH PROFILE

Energy systems researcher working at the intersection of decarbonisation, equity and development, with 5+ years of research experience across India and the United States. Combining macro-energy systems modeling with geospatial analysis and developmental needs to study what it takes to achieve economy-wide decarbonization. Interested in understanding what the different pathways of energy transitions are and their implications on infrastructure development, natural resources, and political economy. Led the net-zero India pathway modelling and sub-national downscaling programme at Princeton's Andlinger Center in collaboration with IIT Delhi. Published on electricity inequality, rural electrification policy, renewable resource assessment and energy design and implementation of energy systems. Experienced in building research partnerships between inter- and trans-disciplinary institutions and in translating modelling results for policy and community stakeholders.

EDUCATION

PhD, Energy Systems and Public Policy 2021

Centre for Technology Alternatives for Rural Areas, Indian Institute of Technology Bombay, Mumbai, India

Thesis: "Understanding Rural Electrification Policy Evolution and Inequality in Residential Electricity Consumption in India" – Received Best PhD Thesis Award

ME, Power and Energy Systems 2015

Bangalore University, Bengaluru, India

Dissertation: "GIS-based Renewable Energy Potential Assessment and Decentralised Energy Planning"

BE, Electrical and Electronics Engineering 2012

Visvesvaraya Technological University (VTU), Karnataka, India

RESEARCH AND PROFESSIONAL EXPERIENCE

Associate Fellow Oct 2023 – Jul 2026

Transitions Research, Goa, India

- Advise on city-level energy transition strategy, including construction of Scope 1, 2 and 3 emissions inventories for small and mid-sized Indian cities.
- Design and pilot emissions-reduction interventions that pair mitigation with community co-benefits.

Schmidt Sciences Visiting Research Scholar Feb 2026 – Jun 2026

School of Public Policy, Indian Institute of Technology Delhi, India

- Extended net-zero India pathway modelling and sub-national downscaling in collaboration with Indian institutional partners.
- Analysed variable renewable energy (VRE) siting across Indian states, identifying land, grid and permitting bottlenecks constraining deployment.
- Conceptualized a community-level VRE siting survey covering prominent RE states to capture local acceptance and land-use conflicts.

Postdoctoral Research Associate Mar 2022 – Dec 2025

Energy Systems Analysis Group, Andlinger Center for Energy and the Environment, Princeton University, NJ, USA

- Led the spatial data collection and analysis for siting VRE infrastructure for the net-zero India modeling and downscaling.
- Built spatially explicit capacity-expansion and siting models of India's power system, integrating land-use, protected-area, and grid-proximity constraints across Indian states.
- Helped in scoping and fostering research collaborations with Indian institutions, developing the project proposal and partnership framework that enabled funding and an MoU for the Net-Zero India Project.
- Master's thesis – 1 (IIT Bombay); Undergraduate students (Princeton): Junior thesis – 2, Senior Thesis – 1; Summer Interns – 4.

Project Coordinating Officer

Sep 2020 – Nov 2021

TUM SEED Centre (Technical University of Munich – IIT Bombay), Mumbai, India

- Coordinated a multi-year, multi-institution living-labs programme testing decentralised energy technologies with rural communities.
- Established the community co-creation and validation protocol used to test and iterate technologies in situ, working across 4 sites and 8 partner institutions.
- A cookstove testbed installed at an NGO, two electric 3-wheelers commissioned for public use and for experiments, and a 5kWp Solar Rooftop System installed at the residential school.

Research Consultant

Nov 2018 – Jun 2019

IIT Bombay–Monash Research Academy, Mumbai, India

- Conducted spatial and national survey analysis of India's electricity access expansion, testing clientelism, incrementalism and maximin as competing explanations of public service delivery.
- Analysis published as a co-authored article in *Energy Research and Social Science* (2021); responsible for dataset construction and analysis.

Project Staff / Research Assistant

Jun 2012 – May 2015

Energy and Wetlands Research Group, Centre for Ecological Sciences, Indian Institute of Science, Bengaluru, India

- Assessed solar, wind and bioenergy potential across 30 districts of Karnataka using remote sensing and GIS, producing resource maps that underpinned 3 peer-reviewed publications and two technical reports.
- Optimised decentralised energy system designs for off-grid and Western Ghats communities, including rooftop PV and standalone energy systems.
- Contributed ecological carrying-capacity assessments informing regional sustainable-development planning.

GRANTS, FELLOWSHIPS AND AWARDS

Schmidt Sciences Visiting Research Fellowship	2026
Competitive visiting fellowship supporting net-zero pathway research at IIT Delhi.	
Best PhD Thesis Award	2022
Centre for Technology Alternatives for Rural Areas, Indian Institute of Technology Bombay.	
Best Poster Award	2017
6th International Conference on Advances in Energy Research (ICAER), IIT Bombay, 12–14 December 2017.	
Sahyadri Young Ecologist Award	2014
LAKE 2014, Energy and Wetlands Research Group, Centre for Ecological Sciences, Indian Institute of Science, Bengaluru.	

PUBLICATIONS

Full list at [Google Scholar](#)

Peer-Reviewed Journal Articles

1. Gwal, S., Sharma, A., Pascale, A., Dasgupta, R., **Hegde, G.**, & Greig, C. (2025). Potential of India's AFOLU sector in achieving net zero: A systematic review of land-based mitigation strategies. *Land Use Policy*, 159, 107798. [\[link\]](#)
2. Sawant, N., & **Hegde, G.** (2022). Missing linkages in the electricity–water nexus in Indian agriculture. *Economic and Political Weekly*, 57(39). [\[link\]](#)
3. Arranz, A. M., Thomson, R., Zech, S., **Hegde, G.**, Arunachalam, D., & Rao, A. B. (2021). The uneven expansion of electricity supply in India: The logics of clientelism, incrementalism and maximin. *Energy Research and Social Science*, 78, 102126. [\[link\]](#)
4. Ramachandra, T. V., & **Hegde, G.** (2015). Energy trajectory in India: Challenges and opportunities for innovation. *Journal of Resources, Energy and Development*, 12(1&2), 1–24. [\[link\]](#)
5. Ramachandra, T. V., **Hegde, G.**, & Krishnadas, G. (2014). Scope for solar energy in Uttara Kannada, Karnataka State, India: Rooftop PV for domestic electricity and standalone systems for irrigation. *Productivity*, 55, 101–119. [\[link\]](#)
6. Ramachandra, T. V., **Hegde, G.**, & Krishnadas, G. (2014). Potential assessment and decentralised applications of wind energy in Uttara Kannada, Karnataka. *International Journal of Renewable Energy Research*, 4(1), 1–10. [\[link\]](#)
7. Ramachandra, T. V., **Hegde, G.**, Setturu, B., & Krishnadas, G. (2014). Bioenergy: A sustainable energy option for rural India. *Advances in Forestry Letters*, 3(1), 1–15. [\[link\]](#)
8. Ramachandra, T. V., & **Hegde, G.** (n.d.). Decentralized renewable energy options for the Western Ghats. *Journal of MGIRE*, 1(1), 24–43. [\[link\]](#)

Manuscripts Under Review and In Preparation

- Anticipating resistance and opportunities in the siting of large-scale renewable energy projects in India
- VRE Siting Challenges In India: An expert elicitation for net-zero Pathways

Book Chapters

1. **Hegde, G.**, Rao, A. B., & Agnihotri, S. B. (2022). Understanding energy inequality in India. In *State of India's Environment Report 2022*. Centre for Science and Environment, New Delhi. [\[link\]](#)
2. **Hegde, G.**, Rao, A. B., & Agnihotri, S. B. (2020). Analysing the inequality pathways of domestic electricity consumption in India. In S. Singh & V. Ramadesigan (Eds.), *Advances in Energy Research*, Vol. 1 (pp. 715–727). Springer Proceedings in Energy, Springer, Singapore. [\[link\]](#)
3. Swami, S., **Hegde, G.**, Rao, A. B., & Agnihotri, S. B. (2020). How do supply-side constraints affect the rural residential feeder parameters? In S. Singh & V. Ramadesigan (Eds.), *Advances in Energy Research*, Vol. 1 (pp. 505–518). Springer Proceedings in Energy, Springer, Singapore. [\[link\]](#)

Conference Papers

1. **Hegde, G.**, Agnihotri, S. B., & Rao, A. B. (2017). Tracing the contours of inequality in household electricity consumption in India. *36th International Energy Workshop (IEW)*, University of Maryland, College Park, MD, 12–14 July 2017. [\[link\]](#)
2. Ramachandra, T. V., & **Hegde, G.** (2014). Scope for distributed renewable energy systems in South India. *IEEE Global Humanitarian Technology Conference – South Asia Satellite*, Kottayam, Kerala, 26–27 September 2014. [\[link\]](#)
3. Ramachandra, T. V., & **Hegde, G.** (2014). Energy latitude in Western Ghats through decentralized renewable energy resources. *SYNERGY with ENERGY: 9th National Conference on the Indian Energy Sector*, Ahmedabad, India, 23–24 May 2014.

Technical Reports

1. Ramachandra, T. V., **Hegde, G.**, & Jain, R. *Solar Energy – The Sustainable Energy Option in Karnataka*. CES Technical Report 132, Energy and Wetlands Research Group, Centre for Ecological Sciences, Indian Institute of Science, Bengaluru. [\[link\]](#)
2. Ramachandra, T. V., Chandran, S. M. D., Joshi, N. V., **Hegde, G.**, & Krishnadas, G. *Sustainable Energy Alternatives for Uttara Kannada*. ENVIS Technical Report 58, Environmental Information System, Centre for Ecological Sciences, Indian Institute of Science, Bengaluru. [\[link\]](#)

PROFESSIONAL SERVICE AND AFFILIATIONS

Peer Review: Energy Policy, Energy Research and Social Science, Applied Energy

TECHNICAL SKILLS

Energy systems modelling: GenX, PIER-RUMI, MACRO, capacity expansion, production cost and downscaling frameworks.

Programming: Julia (JuMP), Python, R, MATLAB, STATA.

Geospatial analysis: QGIS, ArcGIS Pro, FOSS4G stack, remote sensing for renewable resource and land-suitability assessment.

Empirical methods: Participatory research methods, household survey analysis, inequality decomposition, spatial econometrics, systematic review.

Domain: power system operations and planning, SCADA/EMS, grid-connected solar PV, synchrophasor-based smart transmission.

SELECTED PROFESSIONAL TRAINING

Certificate Course in Methods and Applications in Social Science Research (CCMASSR), Institute for Social and Economic Change, Bengaluru | Solar PV Grid-Connected Power Plant, National Training Centre for Solar Technology, KPCL, Bengaluru, 2014 | SCADA/EMS and Power System Protection and Stability, Southern Regional Load Despatch Centre, Bengaluru | Geospatial Analysis with FOSS4G, Indian Institute of Science, Bengaluru, 2014 · DST-SERB School on Smart Transmission Grids using Synchrophasor Technology, IIT Kanpur, 2013.

LANGUAGES

English (fluent), Kannada (native), Hindi (fluent), Marathi (conversational)

REFERENCES

Available on request.